Regularity of the Lorentzian Distance and Length Function at the Singular Boundary in the Kruskal Spacetime

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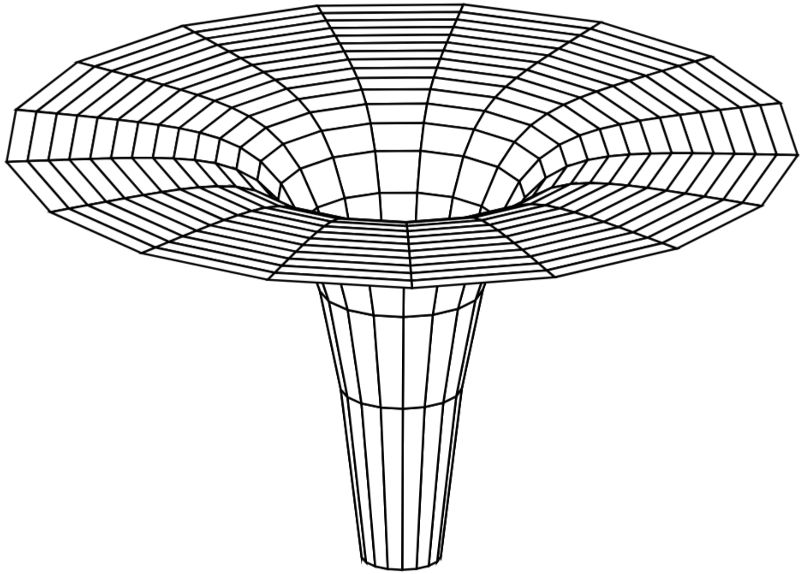
What happens at the singularity?

$$ds^2 = -\left(1 - \frac{r_s}{r}\right)dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$



- Singularities at $r = r_s$ and $r = 0$
- $M_{\text{int}} := (0, r_s) \times \mathbb{R} \times S^2 \Rightarrow \partial_t$ is spacelike
- $M_{\text{ext}} := (r_s, \infty) \times \mathbb{R} \times S^2 \Rightarrow \partial_t$ is timelike

Kruskal Coordinates

- For $r \geq r_s$:

$$T := \left(-1 + \frac{r}{r_s}\right)^{\frac{1}{2}} e^{\frac{r}{2r_s}} \sinh\left(\frac{t}{2r_s}\right), \quad X := \left(-1 + \frac{r}{r_s}\right)^{\frac{1}{2}} e^{\frac{r}{2r_s}} \cosh\left(\frac{t}{2r_s}\right)$$

- For $r \leq r_s$:

$$T := \left(1 - \frac{r}{r_s}\right)^{\frac{1}{2}} e^{\frac{r}{2r_s}} \cosh\left(\frac{t}{2r_s}\right), \quad X := \left(1 - \frac{r}{r_s}\right)^{\frac{1}{2}} e^{\frac{r}{2r_s}} \sinh\left(\frac{t}{2r_s}\right)$$

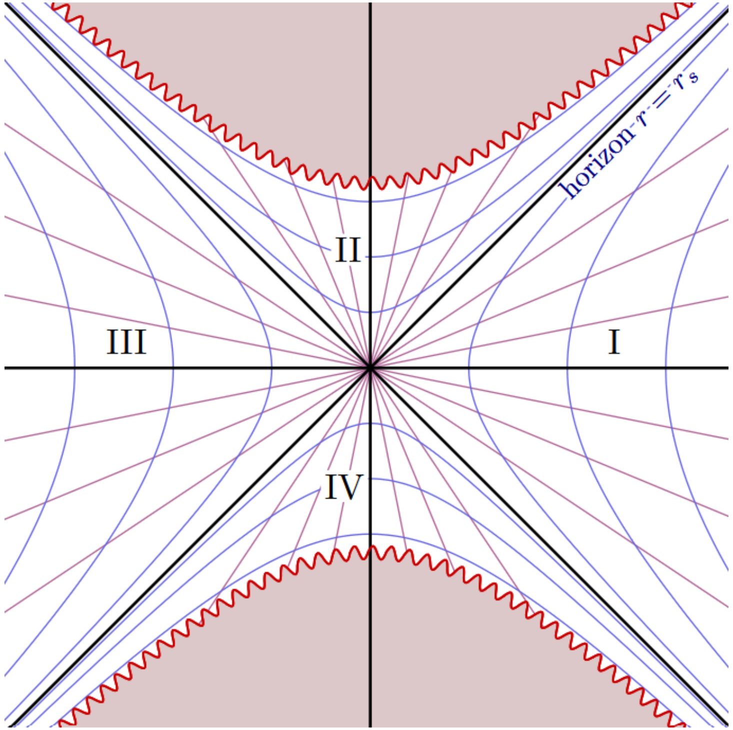
- Metric: $ds^2 = \frac{4r_s^3}{r} e^{-\frac{r}{r_s}} \left(-dT^2 + dX^2\right) + r^2 \left(d\theta^2 + \sin^2\theta d\varphi^2\right)$

Theorem

Kruskal spacetime is isometrically inextendable.

Proof sketch.

Isometric inextendability follows from the divergence of tidal forces along all finite causal geodesics.



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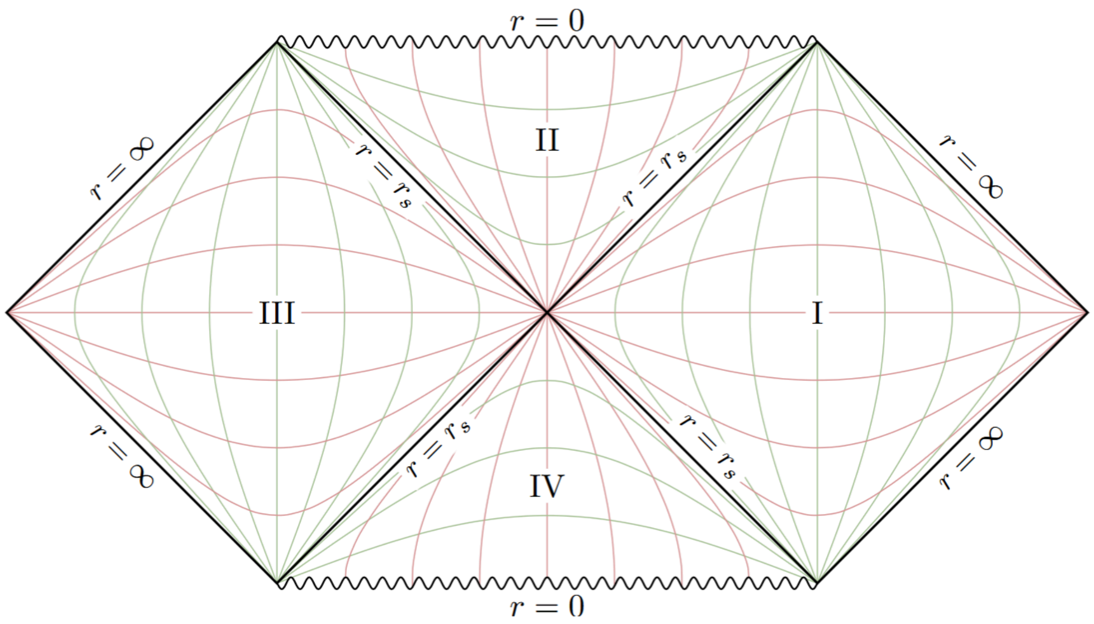


Definition (Lorentzian Lenght and Distance)

$$\ell(\gamma) = \lambda(p, q) := \int_p^q \sqrt{-g(\dot{\gamma}(t), \dot{\gamma}(t))}\, dt$$

$$\tau(p, q) := \sup\{\ell(\gamma) \mid \gamma \text{ is a future-directed causal curve from } p \text{ to } q\}$$

Geometric & Causal Foundation

Definition (Global Hyperbolicity)
A time-oriented Lorentzian manifold (M, g) is called **globally hyperbolic** (g.h.) if it satisfies the following two conditions:

1. (M, g) is causal;
2. $\forall p, q \in M$, the set $J^+(p) \cap J^-(q)$ is compact.

Causal ladder: chronological $\rightarrow$ causal $\rightarrow$ strongly causal $\rightarrow$ *globally hyperbolic*

- M_{ext} and M_{int} are globally hyperbolic when considered separately
- In a g.h. spacetime, for $p \ll q$ there exists a maximizing geodesic realising $\tau(p, q)$
- Killing vectors ∂_t and ∂_φ $\rightarrow$ conserved quantities E an L

$\rightarrow$ Restriction to future-directed timelike geodesics in M_{int}, with r arbitrarily close to 0

Reduction to 1D integrals

$$E^2 = \left(\frac{dr}{d\lambda}\right)^2 + \hbar(r)\left(1 + \frac{L^2}{r^2}\right), \quad \hbar(r) = 1 - \frac{r_s}{r}$$

- Reduction to $\theta = \frac{\pi}{2}$ (by $SO3$-symmetry)
- On M_{int}, r is timelike, so $\frac{dr}{d\lambda} < 0$ along every future-directed timelike curve

$\to$ r strictly monotonic and can be used as a curve parameter $\to$ 1D Integral[1]

$$L = r^2 \frac{d\varphi}{d\lambda}$$

$$\varphi(p,q) = \int_{r_q}^{r_p} \frac{L}{r^2} \frac{dr}{\sqrt{E^2 - \left(1 - \frac{r_s}{r}\right)\left(1 + \frac{L^2}{r^2}\right)}}$$

$$E = \hbar(r) \frac{dt}{d\lambda}$$

$$t(p,q) = \int_{r_q}^{r_p} \frac{E}{1 - \frac{r_s}{r}} \frac{dr}{\sqrt{E^2 - \left(1 - \frac{r_s}{r}\right)\left(1 + \frac{L^2}{r^2}\right)}}$$

[1]Barrett O'Neill, *Semi-Riemannian Geometry : with Applications to Relativity*, 1st ed. (Academic Press, 1983).

The Future Completion

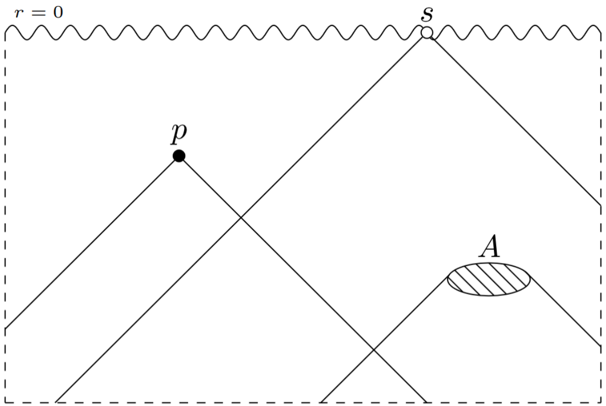
Theorem

The interior $M_{\text{int}} = (r_s, 0) \times \mathbb{R} \times S^2$ admits a future completion that is chronological homeomorphic to $\overline{M}_{\text{int}} = (r_s, 0] \times \mathbb{R} \times S^2$ which includes ideal boundary points.

Proof. The proof can be found in

Topologies on the future causal completion[2]

by Dr. Olaf Müller.



$\square$

[2]Olaf T. Müller, “Topologies on the Future Causal Completion,” 2024, https://arxiv.org/abs/1909.03797.

Model Integral

Proposition

Let $P(x) = ax + bx^2 + cx^3$ with $c \neq 0$. On every interval on which $|P(x)| < 1$, the integral

$$I_\alpha(x) = \int dx\, x^\alpha \,(1 + P(x))^{-\frac{1}{2}}$$

admits a convergent power series representation of the form

$$I_\alpha(x) = \sum_{n,i,j \in Z} \binom{-\frac{1}{2}}{n} \binom{n}{i} \binom{n}{j} c^n \,(-x_1)^{n-i}\,(-x_2)^{n-j}\,\frac{x^{\alpha+n+i+j+1}}{\alpha+n+i+j+1}$$

with

$$x_{1,2} = \frac{-b + -\sqrt{b^2 - 4ac}}{2c}.$$

Proof sketch.

1. Series expansion:

$$(1+u)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} u^n \Rightarrow \text{convergence radius: } |u| < 1$$

2. Factorisation by roots:

$$ax + bx^2 + cx^3 = c(x - x_1)(x - x_2)x$$

3. Expansion of the roots:

$$(x - x_y)^n = \sum_{i=0}^{n} \binom{n}{i} x^i (-x_y)^{n-i}$$

□

Radial Geodesics ($L = 0$)

Energy conditions:

$$E^2 = \left(\frac{dr}{d\lambda}\right)^2 + \left(1 - \frac{r_s}{r}\right)$$

Lorentzian length:

$$\lambda(p, s) = \frac{R^{\frac{3}{2}}}{2\sqrt{r_s}}(\pi - \eta(p) - \sin\eta(p))$$

$$\eta(x) = \arccos\left(\frac{2r_x}{R} - 1\right)$$

$$E = \left(1 - \frac{r_s}{r}\right)\frac{dt}{d\lambda}$$

Time function:

$$t(p, s) = -E\sum_{n,m=0}^{\infty}\binom{-\frac{1}{2}}{n}\frac{r_s^{-n-m-\frac{3}{2}}k^n}{n+m+\frac{5}{2}}r_p^{n+m+\frac{5}{2}}$$

Non-Radial Geodesics ($L \neq 0$)

$$\lambda(p,s) = \frac{1}{L}\sum_{n,i,j}\binom{-\frac{1}{2}}{n}\binom{n}{i}\binom{n}{j}\left(\frac{1}{r_s}\right)^{n-\frac{1}{2}}\frac{(-r_1)^{-i}\,(-r_2)^{-j}}{n+i+j+\frac{5}{2}}\,r_p^{n+i+j+\frac{5}{2}}$$

$$\varphi(p,s) = \sum_{n,i,j}\binom{-\frac{1}{2}}{n}\binom{n}{i}\binom{n}{j}\left(\frac{1}{r_s}\right)^{n+\frac{1}{2}}\frac{(-r_1)^{-i}\,(-r_2)^{-j}}{n+i+j+\frac{1}{2}}\,r_p^{n+i+j+\frac{5}{2}}$$

$$t(p,s) = -\frac{E}{L}\sum_{m,n,i,j}\binom{-\frac{1}{2}}{n}\binom{n}{i}\binom{n}{j}\left(\frac{1}{r_s}\right)^{m+n+\frac{3}{2}}\frac{(-r_1)^{-i}\,(-r_2)^{-j}}{m+n+i+j+\frac{7}{2}}\,r_p^{m+n+i+j+\frac{7}{2}}$$

Hypergeometric Function

$$F_1(a; b_1, b_2; c; x, y) = \sum_{m,n=0}^{\infty} \frac{(a)^{\overline{m+n}} \, (b_1)^{\overline{m}} \, (b_2)^{\overline{n}}}{(c)^{\overline{m+n}} \, m! \, n!} \, x^m \, y^n$$

$$G_1(\alpha; r) := \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \left(\frac{1}{r_s}\right)^{n+\frac{1}{2}} \frac{r^{\alpha+n}}{\alpha+n} F_1\left(\alpha+n; -n, -n; \alpha+n+1; \frac{r}{r_1}, \frac{r}{r_2}\right)$$

$$\lambda(p,s) = \frac{1}{L} G_1\left(\frac{5}{2}; r_p\right) \quad \Bigg| \quad \varphi(p,s) = G_1\left(\frac{1}{2}; r_p\right) \quad \Bigg| \quad t(p,s) = -\frac{E}{L} \sum_{m=0}^{\infty} r_s^{-m-1} \, G_1\left(m + \frac{7}{2}; r_p\right)$$

Regularity at the Singular Boundary

$$\xi : (r_p, E, L) \mapsto (r_p, t(r_p, E, L), \varphi(r_p, E, L))$$

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Proposition

For $r > 0$ satisfying

$$\left| -\frac{r}{r_s} + \frac{r^2}{L^2} + \frac{kr^3}{r_s L^2} \right| < 1,$$

and $E \neq 0$ the Jacobian determinant is non zero, i.e.

$$\det(d\xi) \neq 0.$$

Regularity at the Singular Boundary

Proof sketch.

1. Differentiate explicit power series expressions.
2. Calculate $J = \partial_E \varphi \partial_L t - \partial_L \varphi \partial_E t$
3. Start setting the first five coefficients to be zero and analyze the resulting condition.
4. Result: $r_s = 0$ or $r_1 = r_2 \Rightarrow \Delta = 0$.

$\square$

Regularity at the Singular Boundary

Proposition

Differentiability of λ is ensured also for $L \to 0$ and fixed r_p.

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Proof sketch. Rescale $(r, r_p, L) \mapsto r_s(\varepsilon, \kappa, \delta)$:

1. Differentiate the integral representation and express in relation to:

$$K_\alpha = r_s^{1-\alpha} \int_0^\kappa d\varepsilon \frac{\varepsilon^{\frac{9}{2}-\alpha}}{(k\varepsilon^3 + \varepsilon^2 - \delta^2\varepsilon + \delta^2)^{\frac{3}{2}}}, \quad \alpha \in \{0, 2, 3\}.$$

2. Evaluate $0 \leq \varepsilon \leq \delta$ and $\delta \leq \varepsilon \leq \kappa$ separately.

3. Use the notation

$$f(x,y) \underset{x}{\propto} x^a :\Leftrightarrow \forall y : f(x,y) \in \mathcal{O}(x^a),$$

to find

$$K_\alpha \underset{\delta}{\propto} \delta^{\min\left(\frac{5}{2}-\alpha,0\right)}.$$

4. Use the chain rule with $A = \begin{pmatrix} \partial_E\varphi & \partial_E t \\ \partial_L\varphi & \partial_L t \end{pmatrix}$, and $y^T = (\partial_r\varphi, \partial_r t)$. Inverte it to find no divergence:

$$\begin{pmatrix} \partial_E\lambda \\ \partial_L\lambda \\ \partial_r\lambda \end{pmatrix} = \overbrace{\begin{pmatrix} A & 0 \\ v^T & 1 \end{pmatrix}}^{=d\xi} \begin{pmatrix} \partial_\varphi\lambda \\ \partial_t\lambda \\ \partial_r\lambda \end{pmatrix} \Rightarrow \begin{pmatrix} \partial_\varphi\lambda \\ \partial_t\lambda \\ \partial_r\lambda \end{pmatrix} = \begin{pmatrix} A^{-1} & 0 \\ -v^T A^{-1} & 1 \end{pmatrix} \begin{pmatrix} \partial_E\lambda \\ \partial_L\lambda \\ \partial_r\lambda \end{pmatrix} \underset{\delta}{\propto} \begin{pmatrix} \delta^1 \\ \delta^0 \\ \delta^0 \end{pmatrix}.$$

$\square$

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Differentiability of λ is ensured also for $\left(r_p, L\right) = r_s(\kappa, \delta) \mapsto (0, 0)$.

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Differentiability of λ is ensured also for $\left(r_p, L\right) = r_s(\kappa, \delta) \mapsto (0,0)$.

Proof sketch.

1. Consider the generalized integral

$$K_\alpha = r_s^{1-\alpha} \int_0^\kappa d\varepsilon \frac{\varepsilon^{\frac{9}{2}-\alpha}}{\left(k\varepsilon^3 + \varepsilon^2 - \delta^2\varepsilon + \delta^2\right)^{\frac{3}{2}}},$$

   but this time separatly evaluate $\kappa < \delta$ and $\delta < \kappa$.

2. Same leading order $\to$ introduce $\chi := \max(\kappa, \delta)$ and find

$$K_\alpha \underset{\chi}{\propto} \chi^{\frac{5}{2}-\alpha}$$

by treating $\sigma := \frac{\delta}{\kappa}$ and χ as independant variables.

3. Explicitly determine χ-proportionality of each differential.
4. Use the chain rule again, which takes the form:

$$\begin{pmatrix} \partial_\varphi \lambda \\ \partial_t \lambda \end{pmatrix} = A^{-1} \begin{pmatrix} \partial_E \lambda \\ \partial_L \lambda \end{pmatrix} = \frac{1}{K_0 K_3 + L^2 K_2 K_3} \begin{pmatrix} -E^2 L K_2 K_0 - L K_0 K_3 - L^3 K_2 K_3 \\ E K_0 K_3 + E L^2 K_2 K_3 \end{pmatrix} + \Xi,$$

where Ξ are higher order terms vanishing in the limit. The divergence cancels out. $\square$

Differentiability at the Singular Boundary

Proposition

λ is not differentiable with respect to r.

Proof sketch.

Let $L > 0$ be fixed and $r_p \to 0$. Looking at the integral we know $K_\alpha \underset{\kappa}{\propto} \kappa^{\frac{11}{2}-\alpha}$.

Using the chain rule introduced earlier yields

$$\begin{pmatrix} \partial_\varphi \lambda \\ \partial_t \lambda \end{pmatrix} \underset{\kappa}{\propto} \kappa^{-6} \begin{pmatrix} \kappa^{-\frac{1}{2}} & \kappa^{\frac{5}{2}} \end{pmatrix} \begin{pmatrix} \kappa^{\frac{7}{2}} & \kappa^{\frac{7}{2}} \\ \kappa^{\frac{5}{2}} & \kappa^{\frac{7}{2}} \end{pmatrix} \begin{pmatrix} \kappa^{\frac{11}{2}} \\ \kappa^{\frac{5}{2}} \end{pmatrix} \underset{\kappa}{\propto} \begin{pmatrix} \kappa^{-\frac{1}{2}} \\ \kappa^{\frac{5}{2}} \end{pmatrix}.$$

□

Differentiability at the Singular Boundary

Proposition

Differentiability of λ is also assured with respect to θ.

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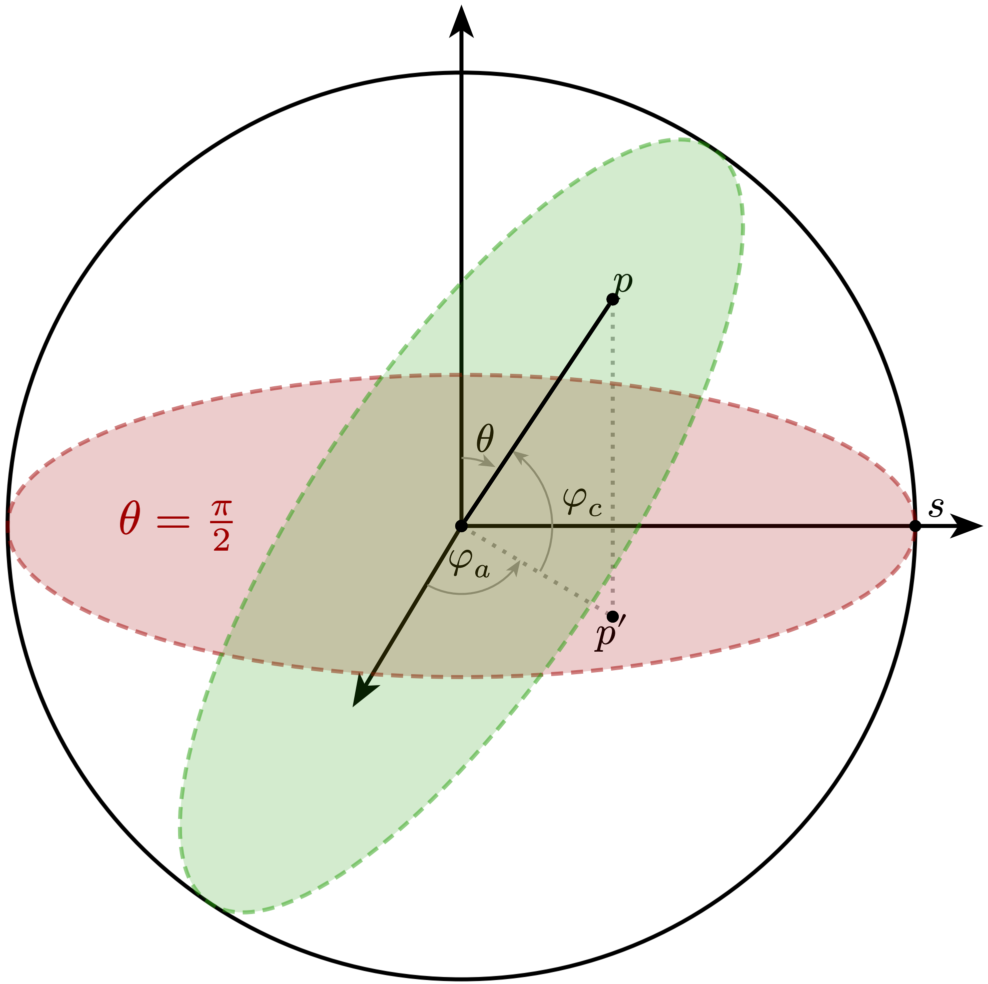
Proof sketch. The angle so far denoted as φ is now φ_c and the actual one is φ_a. The relation between them is given by

$$\cos\varphi_c = \hat{e}_p \cdot \hat{e}_s = \cos\varphi_a \sin\theta.$$

By using the chain rule as before, every divergence cancels out.

The geometric idea:



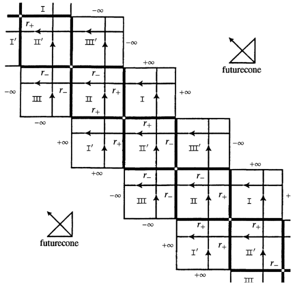
$\square$

Conclusion

We found non-trivial regularity properties at the singular boundaries along ∂_φ and ∂_t.

Next step: Formulate a more generalized statement on the Differentiability of τ with respect to φ and t for any neighbourhood U of $q \in \mathcal{S}$.

$\rightarrow$ Investigate other solutions of the Einstein field equations (i.e. Kerr or Kerr-Newman[3]).



Goal

Find a Synthetic extension to the Kruskal spacetime at $r = 0$.

[3] Barrett O'Neill, *The Geometry of Kerr Black Holes / Barrett O'neill* (Wellesley, Mass., 1995).

[4] This presentation was created using the typst template “taupsi”. Credit and thanks to Lyding Brumm for creating it.

Proof.

Goal

K_m admits no C^2 isometric extension.

Idea: Follow C^0-inextendible future timelike geodesic curves!

Does the geodesic curve $c : (a, b) \to K_m$ cross the horizon $r = r_s$?

Case 1 - yes: $r \circ c(s) \xrightarrow[s \to b]{} 0$

Case 2 - no: $t \circ c(s) \to \infty \Rightarrow \ell(c) = \infty$

Kretschmann Scalar $$\mathcal{K} = R_{ijkl} R^{ijkl} = \frac{12 r_s^2}{r^6} \xrightarrow[r \to 0]{} \infty$$ $\square$

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□

Definition (Chronological Homeomorphism)
A chronological homeomorphism between two spacetimes M and N is a homeomorphism $f : M \to N$. that preserves the chronological relation:

$$p \ll q \Leftrightarrow f(p) \ll f(q).$$

Definition

The Appell series F_1 is defined for $|x| < 1$ and $|y| < 1$ by

$$F_1(a; b_1, b_2; c; x, y) = \sum_{m,n=0}^{\infty} \frac{(a)^{\overline{m+n}} \, (b_1)^{\overline{m}} \, (b_2)^{\overline{n}}}{(c)^{\overline{m+n}} \, m! \, n!} \, x^m \, y^n.$$

Derivatives

$$\frac{\partial^n}{\partial x^n} F_1(a, b_1, b_2, c; x, y) = \frac{(a)^{\overline{n}} (b_1)^{\overline{n}}}{(c)^{\overline{n}}} F_1(a+n, b_1+n, b_2, c+n; x, y)$$

Integral representation

$$F_1(a, b_1, b_2, c; x, y) = \frac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)} \int_0^1 t^{a-1}(1-t)^{c-a-1}(1-xt)^{-b_1}(1-yt)^{-b_2} dt$$

with $\Re c > \Re a > 0$

Finiteness

If we consider Radial geodesics starting from $r_p = r_s$ at rest, we find: $\lambda(p, s) = \frac{\pi r_s}{2}$.

Metric: $g = -\left(1 - \frac{r_s}{r}\right)dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega$

For a future-directed causal curve $\gamma : [r_p, r_q) \to M$ starting at a point p with $\gamma(r_p) = p$, one finds:

$$\lambda(p, q) = \int_{r_q}^{r_p} \sqrt{-g(\dot{\gamma}(r), \dot{\gamma}(r))} dr \leq \int_{r_q}^{r_p} \sqrt{\left(\frac{r_s}{r} - 1\right)^{-1}} dr.$$

Assume $r < \frac{r_s}{2} \Rightarrow \lambda(p, q) < r_p - r_q \leq r_p$.

Continuity: M_{int} is g.h., let $s = (0, \varphi, \theta, t) \in \mathcal{S}$ and $\varepsilon > 0$.

- $U := I^+(\varepsilon, \varphi, \theta, t)$ open nbh. of s
- $|\tau(p, x) - \tau(p, s)| \leq \varepsilon \; \forall x \in U$

Proof sketch. The Jacobi-determinant: $J = \partial_E \varphi \partial_L t - \partial_L \varphi \partial_E t$

$$= \sum_{n,i,j,u,v,k,\ell} \binom{-\frac{1}{2}}{n} \binom{n}{i} \binom{n}{j} \binom{-\frac{1}{2}}{u} \binom{u}{k} \binom{u}{\ell} \frac{E}{L} \left(\frac{1}{r_s}\right)^{u+v+n+2} \frac{r_p^{n+i+j+u+v+k+\ell+4}}{(n+\frac{1}{2}+i+j)(u+\frac{7}{2}+v+k+\ell)}$$

$$\cdot \left[(i\ell - jk) \left(\frac{-2E^2 r_s}{LE(E^2-1)\Delta} \right) - i \cdot \frac{r_s - 2E^2 r_s + 2\Delta}{2LE(E^2-1)\Delta} - j \cdot \frac{-r_s + 2E^2 r_s + 2\Delta}{2LE(E^2-1)\Delta} \right]$$

This is not well-defined for $\Delta \to 0$, but if we take a closer look at its origin:

$$\partial_L \left(r_1^i \, r_2^j \right) = \left[(i-j) \frac{r_s}{2\Delta} + (i+j) \right] \frac{2}{L} k^{i+j-1} \cdot \left(\Delta - \frac{r_s}{2} \right)^{-i} \left(-\Delta - \frac{r_s}{2} \right)^{-j}$$

$\square$

Proof. To get the actual φ_a from the formerly calculated φ_c, we must look at the projection of the point p into the $(\theta = \frac{\pi}{2})$-plane.

Consider

$$\cos\varphi_c = \hat{e}_p \cdot \hat{e}_s = \begin{pmatrix} \cos\varphi_a \sin\theta \\ \sin\varphi_a \sin\theta \\ \cos\theta \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \cos\varphi_a \sin\theta,$$

to find

$$\frac{\partial\varphi_a}{\partial\varphi_c} = \frac{\sin\varphi_c}{\sin\varphi_a \sin\theta}.$$

Now use the chain rule

$$\begin{pmatrix} \partial_\theta \lambda \\ \partial_E \lambda \\ \partial_L \lambda \\ \partial_r \lambda \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & \partial_\theta \varphi_a & 0 & 0 \\ 0 & \partial_E \varphi_a & \partial_E t & 0 \\ 0 & \partial_L \varphi_a & \partial_L t & 0 \\ 0 & \partial_r \varphi_a & \partial_r t & 1 \end{pmatrix}}_{=:B} \begin{pmatrix} \partial_\theta \lambda \\ \partial_{\varphi_a} \lambda \\ \partial_t \lambda \\ \partial_r \lambda \end{pmatrix},$$

and invert B with $x^T := (\partial_\theta \varphi_a, 0)$ to find

$$B^{-1} = \begin{pmatrix} 1 & -x^T A^{-1} & 0 \\ 0 & A^{-1} & 0 \\ 0 & -v^T A^{-1} & 1 \end{pmatrix}.$$

Explicit calculation yields

$$A^{-1} = \frac{1}{J_c} \begin{pmatrix} \xi\, \partial_L t & -\xi\, \partial_E t \\ -\partial_L \varphi_c & \partial_E \varphi_c \end{pmatrix} \quad \text{with} \quad \xi := \frac{\sin\varphi_a \sin\theta}{\sin\varphi_c}$$

$$-\, x^T A^{-1} = \frac{\partial_L t, -\partial_E t}{J_c} \cdot \frac{\cos\varphi_a \cos\theta}{\sin\varphi_c},$$

where $\varphi_c = 0 \Rightarrow \theta = \frac{\pi}{2} \Leftrightarrow \varphi_c = \varphi_a$ $\square$